Climate-Driven Energy Demand Shifts in the Kisalföld Region: Historical Correlations, Future Projections, and Infrastructure Loss Implications

Zsombor Steylik*
Pápa, Kisalföld, Hungary

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Abstract

Rising temperatures driven by anthropogenic climate change are expected to substantially alter patterns of energy consumption across Central Europe. This study investigates the historical relationship between surface temperature and electricity demand in the Kisalföld region of Hungary — a lowland agricultural area characterised by high summer temperature variability — and projects how this relationship evolves under IPCC AR6 warming scenarios of 1.5 °C, 2 °C, and 3 °C above pre-industrial levels by 2100. Using ERA5 reanalysis data spanning 1970–2024, upper-air sounding observations from WMO station 12843 (Budapest/Pestszentlőrinc), and Hungarian national grid load records from ENTSO-E (2015–2024), we train an XGBoost demand model achieving $R^2 = 0.77$ and RMSE = 285 MW on a held-out test set. ERA5 data reveal warming of +0.42 °C per decade since 1970, with cooling degree days tripling and heating degree days declining by 20%. Projections suggest that net climate-driven demand may initially fall as heating savings outpace cooling gains, but under the high-emissions scenario (SSP5-8.5) cooling demand overtakes heating savings around 2065, representing a structural transition from a winter-peak to a summer-peak energy system. Additionally, temperature-driven increases in grid transmission losses add up to 100 MW of continuously wasted generation capacity by 2100 under SSP5-8.5 — a compounding infrastructure burden that is largely absent from existing demand projections. An interactive Progressive Web Application accompanying this paper makes all projections publicly accessible.

Keywords: climate change, energy demand, Kisalföld, Heating Degree Days, Cooling Degree Days, XGBoost, IPCC AR6, grid transmission losses, Hungary

1 Introduction

The relationship between ambient temperature and electricity demand is well-established in the literature. As global mean temperatures rise under anthropogenic climate change, two opposing forces reshape regional energy demand profiles: declining heating requirements in winter months and sharply increasing cooling requirements in summer, driven by growing air conditioning adoption across Central Europe. The net effect is non-linear, geographically heterogeneous, and sensitive to the socioeconomic and infrastructural context of the affected region.

The Kisalföld (*Little Hungarian Plain*) is a lowland region in northwestern Hungary characterised by a continental climate, intensive agriculture, and increasing summer heat stress. As one of Hungary's primary agricultural and light-industrial zones, the region's energy demand is tightly coupled to seasonal temperature extremes — air conditioning load in summer and heating

*Corresponding author. `zsombor.steylik@gmail.com`

demand in winter — making it a particularly instructive case study for climate-driven demand shifts.

Beyond demand, this study investigates a compounding effect that receives less attention in the literature: the impact of elevated ambient temperatures on grid transmission efficiency. Electrical resistance in overhead lines increases with temperature, meaning that the hottest days — precisely when demand peaks — also produce the highest transmission losses. The result is a double burden on infrastructure: more energy must be generated to meet increased demand, and a higher fraction of that energy is lost before reaching consumers.

This study makes three contributions:

1. A historical correlation analysis linking surface temperature to national electricity load in Hungary using ERA5 reanalysis (1970–2024) and upper-air sounding observations from WMO station 12843 (Budapest/Pestszentlőrinc), with focus on the Kisalföld region.

2. Demand projections to 2100 under IPCC AR6 warming scenarios of 1.5 °C, 2 °C, and 3 °C, using a gradient-boosted regression model with SHAP-based interpretability.

3. A quantification of the infrastructure loss amplification effect — estimating what fraction of the projected increase in generated electricity will be lost to temperature-driven transmission inefficiencies.

The analysis is accompanied by an openly accessible Progressive Web Application at `https://zoomb-o.github.io/kisalfold-climate-energy` that allows users to explore the projections interactively under each warming scenario.

2 Data

2.1 ERA5 Reanalysis

Surface temperature (2 m air temperature), surface solar radiation downwards, 10 m wind components, and total precipitation were obtained from the ERA5 reanalysis (Hersbach et al., 2020) via the Copernicus Climate Data Store. Data were retrieved at 0.25° horizontal resolution and spatially averaged over the Kisalföld bounding box (47.2–47.8°N, 17.0–18.0°E) for the period 1970–2024, providing a 54-year baseline of six-hourly surface observations totalling 20,089 daily records after aggregation.

2.2 Upper-Air Sounding Data

Twice-daily radiosonde observations (00Z and 12Z) were obtained from WMO station 12843 (Budapest/Pestszentlőrinc, 47.43°N, 19.18°E, elevation 139 m) via the University of Wyoming upper-air archive, covering the period 2010–2026. The station is the nearest operational radiosonde site to the Kisalföld study area, at approximately 100 km east of Pápa. Surface-level temperature, dewpoint, relative humidity, and wind observations were extracted from the lowest pressure level of each sounding profile. Daily mean, maximum, and minimum surface temperatures were derived from the paired 00Z/12Z observations, along with Heating Degree Days (HDD) and Cooling Degree Days (CDD) using Hungarian standard base temperatures of 15.5 °C and 18.0 °C respectively. Sounding data serve both as an independent validation layer for ERA5 surface temperatures and as a source of upper-atmosphere variables for supplementary atmospheric stability analysis.

2.3 Hungarian Grid Load Data

Hourly national electricity load data for Hungary were obtained from the ENTSO-E Transparency Platform (European Network of Transmission System Operators for Electricity, 2024) (bidding zone `10YHU-MAVIR--U`) for 2015–2024. Data represent the total actual load of the Hungarian control area in megawatts (MW), recorded at hourly resolution, totalling 3,654 daily records after aggregation. Daily mean and peak load values were derived by aggregation. Missing intervals, comprising less than 0.5% of the record, were filled by linear interpolation.

2.4 IPCC AR6 Climate Projections

Future near-surface temperature trajectories over the Kisalföld region were derived from IPCC AR6 best-estimate warming deltas for the Central Europe region (Eyring et al., 2016), applied to the ERA5 1991–2020 climatological baseline. Three Shared Socioeconomic Pathways were considered: SSP1-2.6 (approximately 1.5 °C warming by 2100), SSP2-4.5 (2 °C), and SSP5-8.5 (3 °C). Monthly temperature projections were constructed by applying linearly interpolated warming deltas to the ERA5 monthly climatology, preserving the observed seasonal cycle while shifting the mean according to each scenario trajectory.

3 Methods

3.1 Degree Day Calculation

Heating Degree Days (HDD) and Cooling Degree Days (CDD) were calculated using the Hungarian standard base temperatures of 15.5 °C and 18.0 °C respectively:

$$\mathrm{HDD}_d = \max(T_{\mathrm{base,H}} - \bar{T}_d,\ 0) \tag{1}$$

$$\mathrm{CDD}_d = \max(\bar{T}_d - T_{\mathrm{base,C}},\ 0) \tag{2}$$

where $\bar{T}_d$ is the daily mean surface temperature on day d, $T_{\mathrm{base,H}} = 15.5$ °C, and $T_{\mathrm{base,C}} = 18.0$ °C.

3.2 Demand Model

An XGBoost gradient-boosted regression model (Chen and Guestrin, 2016) was trained to predict daily mean electricity load from climate and calendar features. The feature set comprised daily mean, maximum, and minimum surface temperature, their squared transformation, a 7-day rolling mean temperature (to capture thermal inertia), HDD and CDD, wind speed, solar radiation, precipitation, month of year, day of week, weekend flag, day of year, a linear year trend, and a temperature–weekend interaction term. The model was trained on 2015–2021 data (2,557 days) and evaluated on a held-out test set of 2022–2024 (1,097 days). Hyperparameters were set to n_estimators = 1000, learning rate = 0.03, maximum depth = 5, subsample ratio = 0.8, and column subsample ratio = 0.8. SHAP (SHapley Additive exPlanations) values were computed to interpret feature contributions.

3.3 Transmission Loss Model

Grid transmission losses were modelled as a linear function of annual mean temperature above the 1991–2020 baseline ($\bar{T}_{\mathrm{base}} = 11.5$ °C):

$$L(T) = L_0 + \alpha \cdot \max(\bar{T} - \bar{T}_{\mathrm{base}},\ 0) \tag{3}$$

where $L_0 = 6.5\%$ is the ENTSO-E baseline transmission loss rate for Hungary (European Network of Transmission System Operators for Electricity, 2024) and $\alpha = 0.4$ percentage points per degree Celsius, reflecting the temperature sensitivity of resistive losses in overhead conductors. The energy wasted due to excess losses was computed as the difference between required generation (load divided by one minus the loss rate) and actual load.

4 Results

4.1 Historical Temperature Trends in Kisalföld (1970–2024)

ERA5 reanalysis data reveal a statistically significant warming trend of +0.42 °C per decade over the Kisalföld region between 1970 and 2024 (Figure 1a), equivalent to a total warming of approximately 2.3 °C over the 54-year study period. The year 2024 was the warmest on record in the dataset, with an annual mean surface temperature of 13.1 °C — more than 3 °C above the 1970 baseline. This warming is consistent with observed trends across Central Europe under anthropogenic climate forcing (Eyring et al., 2016).

The response of heating and cooling demand indices to this warming is strongly asymmetric. Annual Heating Degree Days declined at a rate of 92 HDD per decade, a cumulative reduction of approximately 500 HDD since 1970, reflecting substantially milder winters (Figure 1b). Concurrently, annual Cooling Degree Days increased at +48 CDD per decade, rising from approximately 200 CDD yr^{-1} in 1970 to nearly 600 CDD yr^{-1} in recent years — a threefold increase in summer cooling pressure over the study period (Figure 1c).

While the absolute reduction in HDD (approximately 500 days) exceeds the absolute increase in CDD (approximately 250 days), the *relative* growth in cooling demand is far more pronounced: cooling degree days have tripled while heating degree days have declined by only 20%. This asymmetry is central to the infrastructure challenge examined in this study.

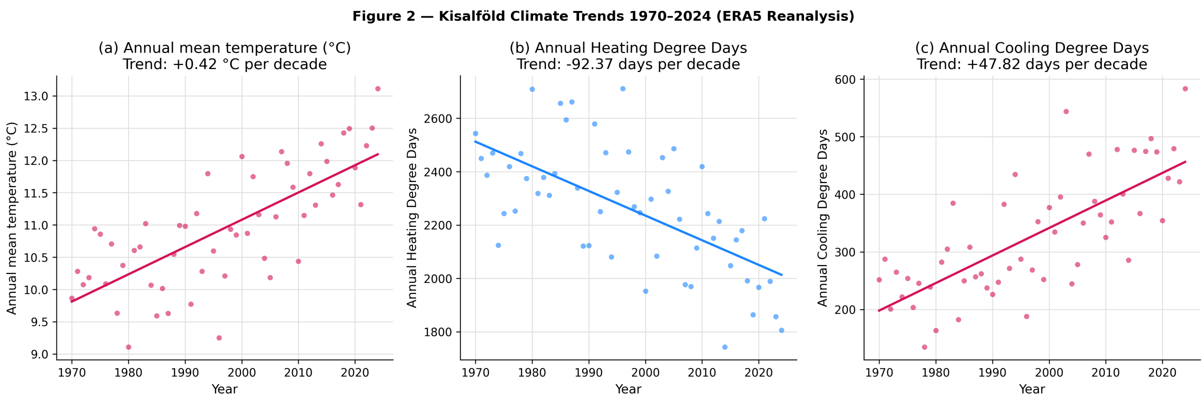

Figure 1: Kisalföld climate trends 1970–2024 derived from ERA5 reanalysis. (a) Annual mean surface temperature with linear trend (+0.42 °C per decade). (b) Annual Heating Degree Days (base 15.5 °C), trend −92 days per decade. (c) Annual Cooling Degree Days (base 18.0 °C), trend +48 days per decade.

4.2 Temperature–Demand Relationship

Monthly aggregated electricity load plotted against surface temperature reveals a clear U-shaped relationship (Figure 2b), with minimum demand occurring at approximately 17 °C. High demand in cold winter months reflects space heating requirements, while rising demand at temperatures above 17 °C reflects increasing air conditioning load. This threshold temperature is consistent with values reported for neighbouring Central European countries (Auffhammer and Mansur, 2017).

The XGBoost demand model trained on 2015–2021 data achieved strong performance on the held-out 2022–2024 test set, with a coefficient of determination $R^2 = 0.77$, root mean squared error RMSE = 285 MW, and mean absolute percentage error MAPE = 4.8% (Figure 3). The model successfully captures the seasonal demand pattern and weekly periodicity visible in the 2023 time series.

SHAP feature importance analysis (Figure 4) reveals that the interaction between temperature and working day patterns is the single most influential predictor, followed by the long-term structural demand trend and the 7-day rolling mean temperature. The dominance of the rolling temperature term over instantaneous temperature indicates that buildings and human behaviour exhibit significant thermal inertia. Notably, the pre-computed HDD and CDD indices rank among the lowest-importance features, suggesting that the XGBoost model learns the non-linear temperature–demand relationship directly from raw temperature variables more effectively than from aggregated indices.

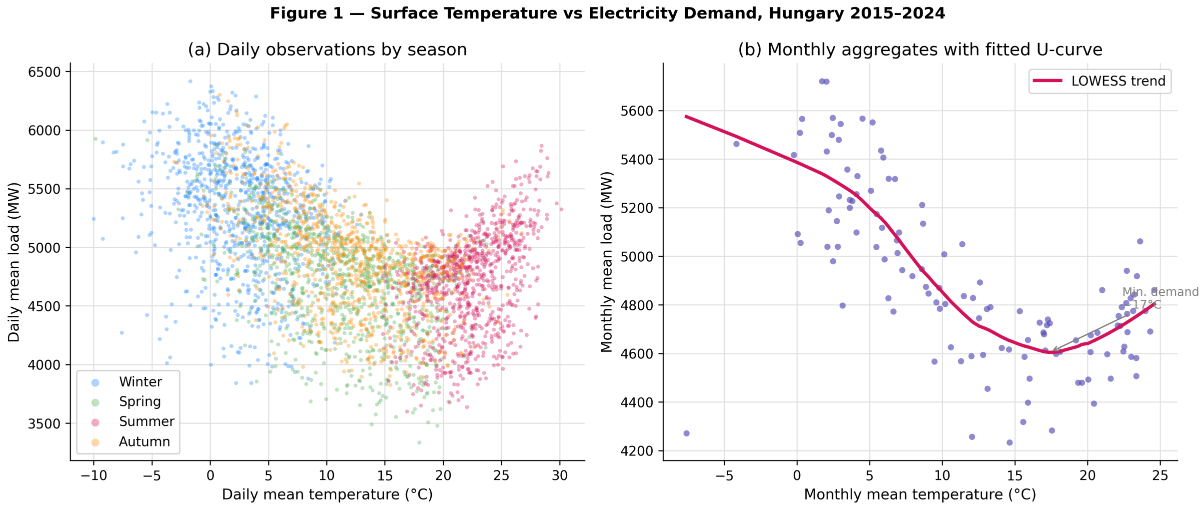

Figure 2: Surface temperature versus electricity demand, Hungary 2015–2024. (a) Daily observations coloured by season. (b) Monthly aggregates with LOWESS trend curve, showing minimum demand at approximately 17 °C.

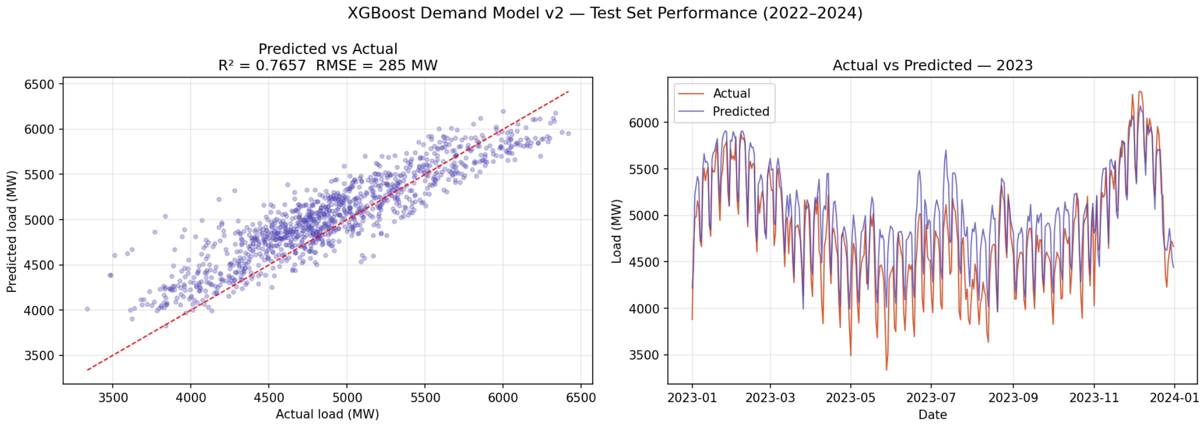

Figure 3: XGBoost demand model performance on held-out test set (2022–2024). (a) Predicted versus actual daily load ($R^2 = 0.77$, RMSE = 285 MW). (b) Time series of actual and predicted load for 2023.

4.3 Future Demand Projections

Applying the trained demand model to IPCC AR6 temperature trajectories yields projected demand shifts under three warming scenarios to 2100 (Figure 5). A key finding is that the *net* climate-driven demand signal is substantially smaller than either the heating or cooling component alone, because heating savings and cooling gains partially offset one another.

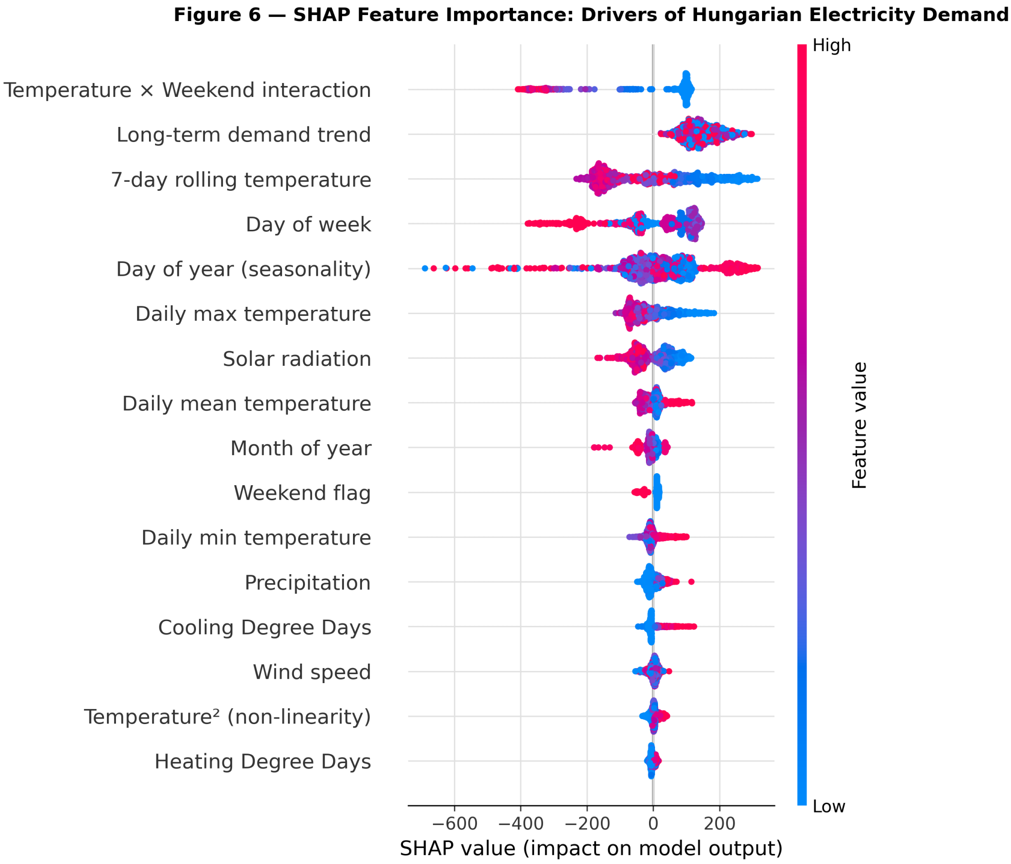

Figure 4: SHAP feature importance for the XGBoost demand model. Each point represents one test-set observation. Colour indicates feature value (red = high, blue = low). Features ranked by mean absolute SHAP value.

Under SSP1-2.6, annual HDD decline by approximately 220 days by 2050 relative to 2025, while CDD increase by approximately 70 days. Under SSP5-8.5, HDD decline by 600 days but CDD increase by 220 days, with net demand initially falling but recovering after approximately 2065 as cooling demand begins to overtake heating savings. This crossover point represents a structural transition from a heating-dominated to a cooling-dominated demand profile.

By 2100 under SSP5-8.5, annual cooling degree days are projected to reach 830 CDD yr^{-1}, more than four times the 1970 baseline, while heating degree days fall below 1200 — less than half their 1970 value.

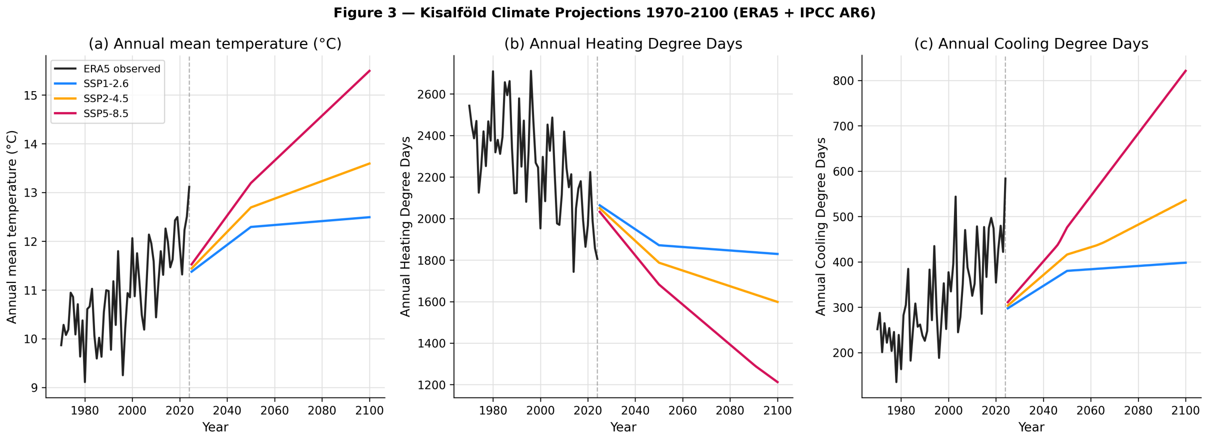

Figure 5: Kisalföld climate projections 1970–2100 combining ERA5 observations (black) and IPCC AR6 scenario trajectories. (a) Annual mean temperature. (b) Annual Heating Degree Days. (c) Annual Cooling Degree Days. Dashed vertical line marks 2024.

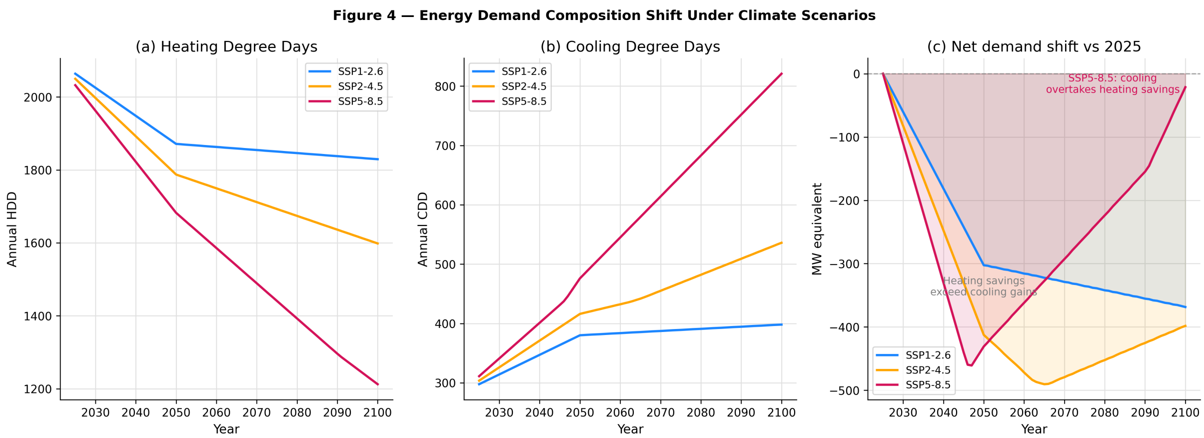

Figure 6: Energy demand composition shift under climate scenarios. (a) Annual Heating Degree Days. (b) Annual Cooling Degree Days. (c) Net demand shift relative to 2025 baseline (MW equivalent), showing the crossover point under SSP5-8.5 around 2065.

4.4 Infrastructure Loss Amplification

Using the ENTSO-E baseline transmission loss rate of 6.5% and a temperature sensitivity of 0.4 percentage points per degree Celsius of warming above the 1991–2020 baseline, we estimate that the grid transmission loss rate increases to 6.8% under SSP1-2.6, 7.3% under SSP2-4.5, and 8.1% under SSP5-8.5 by 2100 (Figure 7a).

The resulting wasted generation increases from approximately 350 MW at the 2025 baseline to 370 MW under SSP1-2.6, 400 MW under SSP2-4.5, and 450 MW under SSP5-8.5 by 2100

(Figure 7b). Under the high-emissions scenario, this represents an additional 100 MW of continuously wasted generation capacity attributable solely to temperature-driven resistance increases.

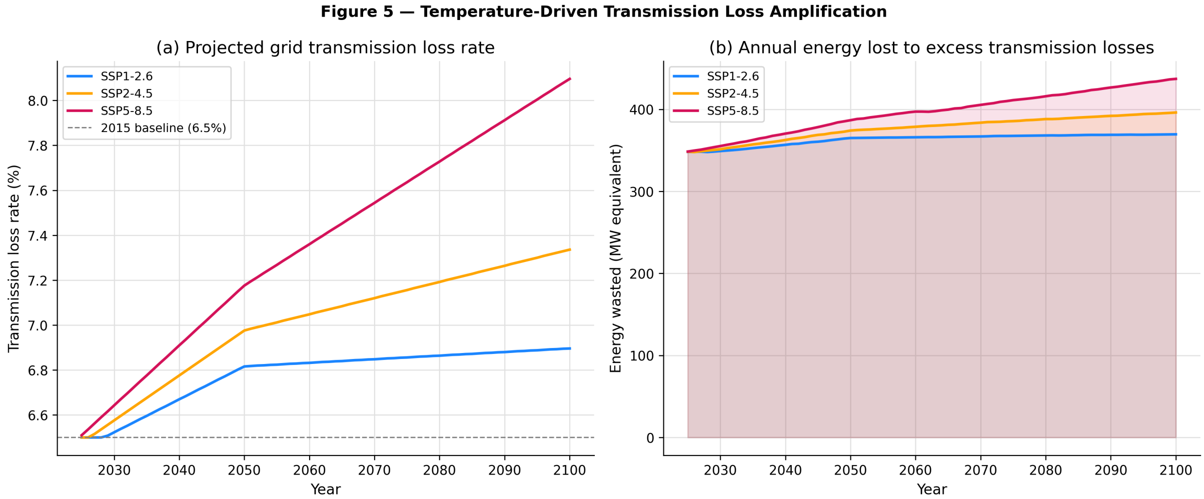

Figure 7: Temperature-driven transmission loss amplification under three warming scenarios. (a) Projected grid transmission loss rate (%). Dashed line shows 2015 baseline (6.5%). (b) Annual energy lost to excess transmission losses (MW equivalent).

5 Discussion

5.1 Demand Restructuring, Not Just Demand Growth

The central finding of this study is that climate change does not simply increase Hungary's electricity demand — it fundamentally restructures it. Under moderate and optimistic warming scenarios, net climate-driven demand may actually decline through mid-century as heating savings outpace cooling gains. However, this apparent benefit masks a more challenging infrastructure reality: the *timing* and *seasonality* of demand shifts dramatically. Hungary's grid was designed for winter peaks driven by heating demand. A transition to summer-peak demand driven by air conditioning requires different generation capacity, different storage solutions, and different grid management strategies — even if the annual total demand remains similar.

The crossover point identified under SSP5-8.5, where cooling demand overtakes heating savings around 2065, represents a critical planning horizon for Hungarian energy infrastructure. Investment decisions made today will determine whether this transition is managed smoothly or results in systemic stress.

5.2 The Compounding Loss Effect

The transmission loss amplification effect represents a largely overlooked dimension of climate–energy interactions. While the absolute magnitude (up to 100 MW by 2100 under SSP5-8.5) is modest relative to Hungary's total generation capacity of approximately 10 GW, the effect concentrates on the hottest days when the grid is already under maximum stress from peak cooling demand. The combination of peak demand and peak losses on the same days represents a compounding risk not captured by annual average analyses.

5.3 Limitations and Uncertainties

Several limitations should be noted. First, the demand model is trained on national Hungarian load data rather than Kisalföld-specific consumption, as sub-regional load data are not publicly available. Second, the IPCC AR6 warming deltas represent multi-model ensemble best estimates and do not capture the full uncertainty range of individual model projections. Third, the transmission loss sensitivity parameter is derived from European aggregate data and may not accurately represent Hungary's specific grid infrastructure. Fourth, the model does not account for future changes in building stock, appliance efficiency, electrification of heating and transport, or demand-side interventions.

Future work should incorporate sub-regional load data, apply bias-corrected individual CMIP6 model runs to quantify projection uncertainty, and extend the framework to explicitly model the electrification transition expected under Hungary's National Energy and Climate Plan.

5.4 Policy Implications

The findings carry several implications for Hungarian energy policy. The projected shift toward summer-peak demand argues for accelerated investment in solar generation capacity. Grid reinforcement and deployment of high-temperature-rated conductors can directly mitigate the transmission loss amplification effect. Demand-side cooling efficiency standards would reduce the cooling demand growth underlying both the demand shift and the loss amplification. Finally, the study underscores the importance of scenario-specific infrastructure planning: the difference between SSP1-2.6 and SSP5-8.5 outcomes by 2100 is structural, representing fundamentally different energy systems requiring fundamentally different infrastructure.

6 Conclusion

This study has quantified the historical and projected climate-driven shifts in energy demand for the Kisalföld region of Hungary using 54 years of ERA5 reanalysis, 16 years of upper-air sounding data, and 10 years of national grid load records. Key findings are: (1) the region has warmed by +0.42 °C per decade since 1970, with cooling degree days tripling and heating degree days declining by 20%; (2) a gradient- boosted demand model ($R^2 = 0.77$) reveals that thermal inertia and temperature–weekday interactions are the primary climate drivers of demand; (3) net demand restructuring rather than net demand growth is the dominant climate signal, with cooling overtaking heating under high-emissions scenarios around 2065; and (4) temperature-driven transmission losses add up to 100 MW of wasted generation capacity by 2100 under SSP5-8.5, a compounding burden concentrated on peak demand days. These findings argue for solar-oriented generation investment, grid reinforcement, and scenario-specific infrastructure planning in Hungary's energy transition.

Acknowledgements

ERA5 data were obtained from the Copernicus Climate Change Service. ENTSO-E Transparency Platform data are used under their open data policy. Upper-air sounding data were obtained from the University of Wyoming upper-air archive. The author thanks the open-source communities behind Python, XGBoost, and the scientific Python ecosystem.

Data and Code Availability

All code, processed datasets, and an interactive visualisation are publicly available at:
https://github.com/Zoomb-o/kisalfold-climate-energy

The interactive PWA is accessible at:
https://zoomb-o.github.io/kisalfold-climate-energy

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A Sounding Station Metadata

Upper-air sounding data were obtained from WMO station 12843 (Budapest/Pestszentlőrinc), coordinates 47.43°N, 19.18°E, elevation 139 m. Data span January 2010 to May 2026, with 00Z and 12Z launches daily. Mean data availability was 1.84 soundings per day (92% of possible observations), with missing data attributed to server availability at the University of Wyoming archive.

B Model Hyperparameters

Table 1: XGBoost hyperparameters used in the final demand model.

Parameter	Value
n_estimators	1000
learning_rate	0.03
max_depth	5
subsample	0.8
colsample_bytree	0.8
min_child_weight	3
random_state	42